

Design and Optimisation of a Polarizer Ring for Precision Energy Calibration of the FCC-ee Collider

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The FCC-ee collider is a proposed electron and positron collider requiring extremely precise beam energy calibration. Accurate determination of beam energy can be achieved using resonant depolarization of polarized electron-positron bunches. While the baseline strategy is to generate polarized bunches directly in the collider, a dedicated lower-energy polarizer ring has been proposed to improve operational efficiency. This project aims to refine and optimise the design of such a polarizer ring through beam dynamics simulations in Xsuite. This will determine key parameters such as ring geometry, lattice configuration, magnet strengths, polarization time, equilibrium polarization, and chromaticity correction.

I. PROJECT AIMS

A. Introduction to the Project Area

The FCC-ee is a proposed 90 km circumference electron-positron collider designed to operate at multiple beam energies. Precision measurements of electroweak observables require extremely accurate beam energy determination. This can be achieved using resonant depolarization of polarized electron-positron bunches.

In storage rings, beams naturally polarize through the Sokolov-Ternov effect due to synchrotron radiation. However, polarization build-up time can be long and may interfere with collider operation. A dedicated lower-energy polarizer ring has therefore been proposed to generate polarized bunches more efficiently.

Initial lattice concepts exist but require refinement in optics design, chromaticity correction, nonlinear dynamics, radiation integrals, and polarization optimization.

B. Description of Aims and Objectives

The overall aim is to design and characterize a viable polarizer ring lattice using numerical simulations.

Objectives include:

- **Lattice Construction:** Developing and testing configurations in Xsuite to determine magnet placement and strengths for dipoles, quadrupoles, and sextupoles.
- **Evaluation of Realistic Properties:** Assessing a ring with realistic imperfections, such as alignment errors and unwanted magnet multipolar components, and developing their corrections.
 - Studying nonlinear beam dynamics and dynamic aperture.
 - Evaluating momentum acceptance through chromaticity correction

- **Polarization Performance:** Estimating polarization build-up time and equilibrium polarization. *Note: These studies will be conducted alongside closed orbit perturbations, as polarization results are only meaningful in the presence of such effects.*
- **Outcome:** A technically documented lattice proposal with quantified performance characteristics.

C. Motivation and Beneficiaries

The project is of interest to the FCC-ee collaboration, with the potential for the design to become a baseline concept for further accelerator development. A dedicated polarizer ring would increase efficiency of collider operation by separating polarization generation from physics data-taking.

II. PROJECT PLAN

A. Lattice Design and Simulation

The work will be carried out primarily using Xsuite. A baseline lattice will be constructed and analysed to determine Twiss parameters and tunes.

B. Chromaticity and Nonlinear Optimisation

Sextupoles will be implemented to correct chromaticity and increase momentum acceptance. Tracking simulations will assess dynamic aperture. Misalignments and their subsequent corrections will be implemented early in the schedule, immediately following initial dynamic aperture and momentum acceptance studies.

C. Radiation and Polarization Studies

Synchrotron radiation effects will be included to estimate polarization build-up time and equilibrium polarization. Where feasible, spin tracking tools in Xsuite will

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be used to evaluate performance, ensuring that orbit perturbations are factored in to maintain physical relevance.

III. PROJECT DEPENDENCIES AND TRAINING NEEDS

The successful execution of this project depends on the understanding of the following concepts:

- General accelerator physics and beam dynamics.
- Specialised domains such as synchrotron radiation and polarization mechanisms.
- Proficiency in Xsuite and access to sufficient computational resources.

IV. RISK REGISTER

- **Technical obstacles:** Potential for insufficient dynamic aperture, excessive chromaticity, or long polarization build-up times.
- **Operational Integration:** A risk exists that, contrary to initial studies, sufficient polarization cannot be maintained during the transfer/injection into the main collider, as this has not yet been studied.
- **Mitigation:** Risks will be mitigated through iterative lattice optimization and sensitivity studies.